

COFFEE PACKAGING QUALITY INSPECTION REPORT

Real-Time AI Seal Inspection and Physical Leak Detection

1. Report Information

Packet ID	PKT-20260829232840-84C7
Report ID	RPT-20260829-E357DB
Inspection Date	29 August 2026
Inspection Time	11:31:45 PM
Inspection Module	Packet Seal and Leak Quality Inspection

2. Overall Inspection Summary

Inspection	Result
Real-Time AI Overheat Inspection	OVERHEAT_DETECTED
Physical Packet Leak Test	LEAK
Overall Quality Decision	REJECT

3. Final Quality Decision

REJECT

Potential overheat defect was detected by the real-time AI inspection system.

4. Real-Time Two-Stage AI Inspection

Detected Seal Regions	3
Overheat Detected	YES
Highest Overheat Confidence	47.6%
AI Inspection Status	OVERHEAT_DETECTED

Seal-by-Seal Results

Seal	Seal Confidence	Overheat	Overheat Confidence
Seal 1	81.5%	NO	-
Seal 2	48.0%	NO	-
Seal 3	34.1%	NO	-

PACKET: OVERHEAT DETECTED

SEAL: R1 4% | OVERHEAT

overheat 47.6%

overheat 29.9%

overheat 33.4%

SEAL: R4 1% | NO_OVERHEAT DETECTED

5. Physical Packet Leak Detection

Leak Test Result	LEAK
Initial Value	281533
Average Value	277038.68
Threshold	220000
Minimum	272437
Maximum	278828
Range	6391
Reading Count	10
Final Device Status	TOP_REACHED

Load Cell Readings

Reading Number	Sensor Value
1	272437
2	274018
3	276299
4	277254
5	278049
6	278140
7	278074
8	278602
9	278686
10	278828

6. Recommended Action

- Hold or reject the affected packet.
- Inspect sealing temperature settings.
- Verify sealing dwell time and heater condition.
- Inspect subsequent packets from the same batch.
- Reject the leaking packet.
- Inspect sealing pressure and seal alignment.
- Check the packaging material for damage.

7. Inspection System Information

AI Stage 1	Seal Region Object Detection
AI Stage 2	Overheat Defect Object Detection
Camera	IP Webcam Real-Time Video
Physical Controller	Arduino Uno
Force Measurement	HX711 Load Cell System
Mechanical Test	Motorized Packet Pressure Mechanism
Safety / Position Control	Top and Bottom Limit Switches

This report was generated automatically using the coffee packet quality inspection system.